

Assignment 1

Blockchain Technologies

Tasks

1. Write a smart contract
 - Smart contract should be able to receive Ether token
 - Smart contract's owner should be able to withdraw all the Ether in smart contract balance
 - Smart contract should have a functionality to check the balance of the contract
2. Configure environment to run web3.js and install library web3.js
3. Configure web3.js to run with Ganache and/or any Public Testnet
4. Configure Metamask with Ganache and/or any Public Testnet
5. Deploy the contract to Ganache and/or any Public Testnet
6. Call any functions of smart contract from step 4 from environment which you have configured from steps 1,2,3

Project Repository structure

- README.md
 - - Title
 - Usage
 - Put here demo screenshots (png or gif)
 - Examples
- LICENSE
 - For
example: [https://github.com/OpenZeppelin/contracts/blob/master/LICENSE](https://github.com/OpenZeppelin/openzeppelin-contracts/blob/master/LICENSE)

Create repository in github, push it into the github and submit the link to the repository into the moodle. Assignment will not be accepted if Repository and README are not submitted and structured properly!!!

References

- 1. <https://web3py.readthedocs.io/en/stable/quickstart.html>
- 2. <https://web3js.readthedocs.io/en/v1.10.0/getting-started.html>
- 3. <https://trufflesuite.com/ganache/>
- 4. <https://www.youtube.com/watch?v=t3wM5903ty0&list=PLS5SEs8ZftgXlCGXNfzKdq7nGBclaVOdN>
- 5. <https://medium.com/@kacharlabhargav21/using-ganache-with-remix-and-metamask-446fe5748ccf>

Grading

- Submit in groups
 - Group can have 1,2 or 3 person

Category	Points
Smart Contract Development	30
Deployment	40
Deployment to any public Testnet	25
Deployment to Ganache	15
Integration with Web3.js	20
Readme	10